



< Previous













Next >

5.4.1 Hermitian Positive Definite matrices

 Bookmark this page

Pursue a verified certificate

Verified Certificate

 THE UNIVERSITY of TEXAS SYSTEM

This is to certify that

Sandipan Dey

successfully completed and received a passing grade in

UT.ALA:Advanced Linear Algebra: Foundations to Frontiers

a course of study offered by UTAustinX, an online learning initiative of University of Texas System.

**Maurilio Pugliesi**
Professor
UniversityX

**Justine Doe, Ph.D.**
Professor
UniversityX

 Verified Certificate

Valid Certificate ID

Issued Month / Year

12345678901234567890

- Unlimited access to all course materials
- Shareable [verified certificate](#)
- AI learning assistant and translations

Upgrade for \$99

Video

▶ 5:02 / 5:02

▶ 2.0x

🔊

🔌

CC

“

equal to chi Hermitian
alpha chi, is greater
than zero. Okay?
Well, what is the
generalization of
that? If

you're given the
matrix A then you can
look at psi is equal to
x Hermitian A
times x. This
obviously is a scalar.
Right? Now if A is
symmetric and

positive-definite.
Let's take the case
where x only has two
entries and

therefore A is 2 by 2.
if you have a situation
where x not being the
zero

vector automatically

Video

📄 Download video file

Transcripts

📄 Download SubRip (.srt) file

📄 Download Text (.txt) file

Reading assignment

1/1 point (graded)

Read Unit 5.4.1 of the notes. [\[LINK\]](#)

☒ done

✓

Submit

✓ Correct (1/1 point)

Discussion

Hide Discussion

Topic: Week 05 / 05.4.1

Add a Post

Show all posts ▼

by recent activity ▼

There are no posts in this topic yet.

✕

Homework 5.4.1.1

1/1 point (graded)
Let $\mathbf{B} \in \mathbb{C}^{m \times n}$ have linearly independent columns.

$\mathbf{A} = \mathbf{B}^H \mathbf{B}$ is HPD.

ALWAYS

✔ Answer: ALWAYS

Prove it!

Solution:

For a complete solution, see [the notes](#).

Submit

📘 Answers are displayed within the problem

Homework 5.4.1.2

1/1 point (graded)
Let $\mathbf{A} \in \mathbb{C}^{m \times m}$ be HPD.

The diagonal elements of $\mathbf{A}$ are real and positive.

ALWAYS

✔ Answer: ALWAYS

Prove it!

Solution:

For a complete solution, see [the notes](#).

Submit

📘 Answers are displayed within the problem

Homework 5.4.1.3.

1/1 point (graded)
Let $\mathbf{A} \in \mathbb{C}^{m \times m}$ be HPD. Partition

$$\mathbf{A} = \left(\begin{array}{c|c} \alpha_{11} & \mathbf{a}_{21}^H \\ \hline \mathbf{a}_{21} & \mathbf{A}_{22} \end{array} \right).$$

$\mathbf{A}_{22}$ is HPD.

ALWAYS

✔ Answer: ALWAYS

Prove it!

Solution:

For a complete solution, see [the notes](#).

Submit

i Answers are displayed within the problem

< Previous

Next >

© All Rights Reserved



edX

- [About](#)
- [Affiliates](#)
- [edX for Business](#)
- [Open edX](#)
- [Careers](#)
- [News](#)

Legal

- [Terms of Service & Honor Code](#)
- [Privacy Policy](#)
- [Accessibility Policy](#)
- [Trademark Policy](#)
- [Sitemap](#)
- [Cookie Policy](#)
- [Your Privacy Choices](#)

Connect

- [Idea Hub](#)
- [Contact Us](#)
- [Help Center](#)
- [Security](#)
- [Media Kit](#)

